```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack> g++ stack.c -o stack.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack> ./stack.exe

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:1
Enter value:4
4 pushed.

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:1
Enter value:5
5 pushed.

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:1
Enter value:6
Stack Overflow!
```

Windows PowerShell

```
1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:3
Top element is 5

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:4
Stack full

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:5
Stack not empty

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:2
Popped 5
```

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Windows PowerShell

```
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:5
Stack not empty

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:2
Popped 5

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:6
Stack elements:
4

1.Push
2.Pop
3.Peek
4.Isfull
5.Isempty
6.Show
(Any other key to exit)
Enter choice:/
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sta
```

```
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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack> g++ postpre.c -o postpre.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack> ./postpre.exe

1.Postfix
2.Prefix
Any other key to Exit
Choice:1
Enter expression:A$B*C-D+E/F/(G+H)
Postfix: AB$C*D-EF/GH+/+

1.Postfix
2.Prefix
Any other key to Exit
Choice:2
Enter expression:A$B*C-D+E/F/(G+H)
Prefix: -*$ABC+D/E/F+GH

1.Postfix
2.Prefix
Any other key to Exit
Choice:/
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack>
```

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```
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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack> g++ evaluation.c -o evaluation.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack> ./evaluation.exe
1. Postfix
2. Prefix
Any other key to Exit
Enter choice: 1
Enter expression: 623+-382/+*2$3+
Result = 52.00

1. Postfix
2. Prefix
Any other key to Exit
Enter choice: 2
Enter expression: +-/6312
Result = 3.00

1. Postfix
2. Prefix
Any other key to Exit
Enter choice: /
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Stack>
```

Windows PowerShell

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Queue> g++ linear_queue.c -o linear_queue.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Queue> ./linear_queue.exe
Choose operation:
1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: 1
Enter a number: 4
4 enqueued.

1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: 1
Enter a number: 5
5 enqueued.

1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: 4
Queue full!!

1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: 2
4 Dequeued

1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: 2
5 Dequeued

1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: 3
Queue empty!!

1.Enqueue
2.Dequeue
3.Check if Queue empty
4.Check if Queue full
Enter your choice: /
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Queue>
```

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```
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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Queue> g++ circular_queue.c -o circular_queue.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Queue> .\circular_queue.exe
Choose operation:
1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 3
Queue empty!!

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 1
Enter a number:4
4 enqueued.

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 1
Enter a number:5
5 enqueued.

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 4
Queue full!!

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 2
4 Dequeued

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 1
Enter a number:6
6 enqueued.

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: 5
QUEUE elements
5
6

1.Enqueue
2.Dequeue
3.Check if empty
4.Check if full
5.Show
Enter choice: /
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Queue>
```

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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
 g++ factorial.c -o factorial.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
 .\factorial.exe
Enter a number:4
4!=24
Enter a number:5
5!=120
Enter a number:6
6!=720
Enter a number:7
7!=5040
Enter a number:8
8!=40320
Enter a number:.
program terminated..
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
```

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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion> g++ fibonacci.c -o fibonacci.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion> .\fibonacci.exe

Enter no of terms:4
0       1       1       2
Enter no of terms:7
0       1       1       2       3       5       8
Enter no of terms:10
0       1       1       2       3       5       8       13      21      34
Enter no of terms:12
0       1       1       2       3       5       8       13      21      34      55      89
Enter no of terms:/
Program terminated!!
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
```

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```
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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
g++ GCD.c -o GCD.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
.\GCD.exe
Enter two numbers:5 7
GCD=1
Enter two numbers:8 40
GCD=8
Enter two numbers:10 35
GCD=5
Enter two numbers:/
Porgram terminated...
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
g++ tower_of_hanoi.c -o tower_of_hanoi.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Recursion>
.\tower_of_hanoi.exe
Enter N:3
Move 1 from A to C
Move 2 from A to B
Move 1 from C to B
Move 3 from A to C
Move 1 from B to A
Move 2 from B to C
Move 1 from A to C
Enter N:4
Move 1 from A to B
Move 2 from A to C
Move 1 from B to C
Move 3 from A to B
Move 1 from C to A
Move 2 from C to B
Move 1 from A to B
Move 4 from A to C
Move 1 from B to C
Move 2 from B to A
Move 1 from C to A
Move 3 from B to C
Move 1 from A to B
Move 2 from A to C
Move 1 from B to C
Enter N:/
Program terminated...
```

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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
g++ singlelinklist.c -o singlelinklist.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
.\singlelinklist.exe

--- Linked List Menu ---
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice:1

1)Beginning
2)In-between
3)Last
Choice:1
Enter a value:5

--- Linked List Menu ---
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice:1

1)Beginning
2)In-between
3)Last
Choice:1
Enter a value:4

--- Linked List Menu ---
1)Add Node
2)Remove Node
```

```
2)Remove Node
3)Display
4)Exit
Choice:1

1)Beginning
2)In-between
3)Last
Choice:3
Enter a value:6

--- Linked List Menu ---
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice:2

1)Beginning
2)In-between
3)Last
Choice:2
Enter the position of node to remove:2
Removed:5

--- Linked List Menu ---
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice:3
LIST:4->6->NULL

--- Linked List Menu ---
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice:4
Exiting...
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
 g++ double_link_list.c -o double_link_list.exe
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
.\double_link_list.exe

1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 1

1)Beginning
2)In-between
3)Last
Choice: 1
Enter a value: 6

1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 1

1)Beginning
2)In-between
3)Last
Choice: 3
Enter a value: 8

1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 1

1)Beginning
2)In-between
3)Last
Choice: 2
Enter position(1-based)and value: 2
7

1)Add Node
2)Remove Node
3)Display
```

```
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 3
LIST(Forward):  6 <-> 7 <-> 8 <-> NULL

1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 2

1)Beginning
2)In-between
3)Last
Choice: 2
Enter the position of node to remove: 2
Removed: 7
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 2

1)Beginning
2)In-between
3)Last
Choice: 3
Removed: 8
1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: 3
LIST(Forward):  6 <-> NULL

1)Add Node
2)Remove Node
3)Display
4)Exit
Choice: /
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
```

Windows PowerShell

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List> g++
 .\circular_linked_list.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List> .\c
ircular_linked_list.exe
1) Create/Start List
2) Exit
Choice: 1
Enter a value: 5

MAIN MENU:
1)Add
2)Remove
3)Display
4)Exit
Choice: 1

--- ADD MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 1
Enter a value: 4

--- ADD MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 3
Enter a value: 8

--- ADD MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 2
Enter position (1-based) and value: 3 7

--- ADD MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 4

MAIN MENU:
1)Add
2)Remove
3)Display
4)Exit
Choice: 3
LIST: 4 -> 5 -> 7 -> 8 -> (head)
```

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```
MAIN MENU:
1)Add
2)Remove
3)Display
4)Exit
Choice: 2

--- REMOVE MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 1
Removed: 4

--- REMOVE MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 3
Removed: 8

--- REMOVE MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 2
Enter the position of node to remove: 2
Removed: 7

--- REMOVE MENU ---
1) Beginning
2) Between
3) Last
4) Back
Choice: 4

MAIN MENU:
1)Add
2)Remove
3)Display
4)Exit
Choice: 3
LIST: 5 -> (head)

MAIN MENU:
1)Add
2)Remove
3)Display
4)Exit
Choice: 4
Exiting...
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
```

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Windows PowerShell

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List> g++ .\link_stack.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List> .\link_stack.exe

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 1
Enter value: 1
1 pushed to stack.

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 1
Enter value: 2
2 pushed to stack.

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 1
Enter value: 3
3 pushed to stack.

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 4
STACK: 3 -> 2 -> 1 -> NULL

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 2
Popped Value: 3

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 3
Top Value: 2

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 4
STACK: 2 -> 1 -> NULL

1)Push
2)Pop
3)Seek
4)Display
5)Exit
Choice: 5
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
```

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Windows PowerShell

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List> g++ list_queue.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List> .\list_queue.exe
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 1
Enter value to enqueue: 1
1 added to queue.
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 1
Enter value to enqueue: 2
2 added to queue.
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 1
Enter value to enqueue: 3
3 added to queue.
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 4
QUEUE (Front to Rear): 1 -> 2 -> 3 -> NULL
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 3
Front Value: 1
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 2
Dequeued Value: 1
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 2
Dequeued Value: 2
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 4
QUEUE (Front to Rear): 3 -> NULL
1) Enqueue
2) Dequeue
3) Peek
4) Display
5) Exit
Choice: 5
Exiting...
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\List>
```

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```
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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\bubblesort.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\bubblesort.exe
Enter no of terms:10
Enter values:3 84 31 49 72 57 26 43 58 17
3       17      26      31      43      49      57      58      72      84
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\Selection_sort.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\Selection_sort.exe
Enter no of terms:10
Enter values:73 31 37 38 18 98 91 39 87 50
18      31      37      38      39      50      73      87      91      98
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\insertion.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\insertion.exe
Enter no of terms:10
Enter values:17 60 16 45 39 6 41 31 12 18
6       12      16      17      18      31      39      41      45      60
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\shell_sort.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\shell_sort.exe
Enter the size of array:10
Enter values:96 18 1 14 100 54 58 47 83 84
Sorted array!!
1       14      18      47      54      58      83      84      96      100
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\merge.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\merge.exe
Enter no of terms:10
Enter values:52 38 62 57 47 35 100 50 9 77
9       35      38      47      50      52      57      62      77      100
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\heap.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\heap.exe
Enter no of terms:10
Enter values:51 68 92 50 46 96 69 24 82 4
4       24      46      50      51      68      69      82      92      96
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> g++ .\quick.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort> .\quick.exe
Enter size of array : 10
Enter elements in array : 40 5 61 47 3 39 30 45 58 43
Sorted elements in array : 3    5       30      39      40      43      45      47      58      61
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Sort>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> g++ .\separate_hashing.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> .\separate_hashing.exe
Enter the number of available locations (hash table size): 10
Enter numbers:
8 88 53 3 47 6 69 40 60 99 9 43 88 74 86 10 71 86 82 57 /

--- Hash Table Contents ---
Index [0]: 40 -> 60 -> 10 -> NULL
Index [1]: 71 -> NULL
Index [2]: 82 -> NULL
Index [3]: 53 -> 3 -> 43 -> NULL
Index [4]: 74 -> NULL
Index [5]: NULL
Index [6]: 6 -> 86 -> 86 -> NULL
Index [7]: 47 -> 57 -> NULL
Index [8]: 8 -> 88 -> 88 -> NULL
Index [9]: 69 -> 99 -> 9 -> NULL
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing>
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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\search>
 g++ .\linear_search.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\search>
 .\linear_search.exe
Enter no of terms:10
Enter values:74 28 84 48 87 39 45 60 85 46
Enter the no to find:87
Found at position 5
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\search>
```

```
Windows PowerShell
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\search>
 g++ .\binary_search.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\search>
 .\binary_search.exe
Enter no of terms:10
Enter values:57 46 8 71 74 86 50 14 15 66
Enter the no to find:50
sorted array
8       14      15      46      50      57      66      7
1      74      86
Found at position 5
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\search>
```

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PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> g++ .\linear_hashing.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> .\linear_hashing.exe
Enter the number of available location:10
Enter numbers:84 19 85 80 30 1 30 78 39 47
80      30      1       30      84      85      39      47      78      19
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> g++ .\quadratic_hashing.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> .\quadratic_hashing.exe
Enter the number of available location:10
Enter numbers:47 50 35 94 74 43 41 81 69 20
50      41      81      43      94      35      20      47      74      69
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> g++ .\double_hashing.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing> .\double_hashing.exe
Enter the number of available locations:10
Enter numbers: 24 16 9 38 12 4 34 86 20 8

Final Hash Table:
20      8       12      34      24      4       16      86      38      9
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Hashing>
```

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> g++ .\Tree_traversal.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> .\Tree_traversal.exe
Enter integers (enter any non-numeric character to stop): 59 3 85 11 87 42 39 33 59 6 27 14 44 /

1) Inorder
2) Postorder
3) Preorder
4) Back
Choice: 1
Inorder: 33     11      59      3       6       87      27      59      14      42      44      85      39

1) Inorder
2) Postorder
3) Preorder
4) Back
Choice: 2
Postorder: 33   59      11      6       27      87      3       14      44      42      39      85      59

1) Inorder
2) Postorder
3) Preorder
4) Back
Choice: 3
Preorder: 59    3       11      33      59      87      6       27      85      42      14      44      39

1) Inorder
2) Postorder
3) Preorder
4) Back
Choice: 4

Enter 'Y' for new session: n
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree>
```

Windows PowerShell

```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> g++ .\binary_search_tree.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> .\binary_search_tree.exe
Enter integers for BST (non-numeric to stop): 14 17 11 7 53 4 13 12 /

1) Display Inorder
2) Search for Value
3) Exit Session
Choice: 1
Inorder: 4      7       11      12      13      14      17      53

1) Display Inorder
2) Search for Value
3) Exit Session
Choice: 2
Enter value to search:
12
Search Path: 14 -> 11 -> 13 -> 12
Result: Value 12 found in the tree!

1) Display Inorder
2) Search for Value
3) Exit Session
Choice: 2
Enter value to search: 6
Search Path: 14 -> 11 -> 7 -> 4 -> NULL
Result: Value 6 not found in the tree.

1) Display Inorder
2) Search for Value
3) Exit Session
Choice: 3

New tree session? (Y/N): n
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> g++ .\AVL.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> .\AVL.exe
Enter integers for AVL tree (non-numeric to stop): 14 17 11 7 53 4 13 12 8 6
0 19 16 20 /

AVL Tree (Inorder with Heights): 4 (H:1)
7 (H:2)
8 (H:1)
11 (H:3)
12 (H:2)
13 (H:1)
14 (H:4)
16 (H:1)
17 (H:2)
19 (H:3)
20 (H:1)
53 (H:2)
60 (H:1)

PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> g++ .\prim_min_tree.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> .\prim_min_tree.exe
Enter number of vertices (max 20): 8
Enter number of edges: 15
Enter edge 1 (source destination weight): 0 5 3
Enter edge 2 (source destination weight): 0 4 3
Enter edge 3 (source destination weight): 0 6 5
Enter edge 4 (source destination weight): 0 1 6
Enter edge 5 (source destination weight): 0 7 8
Enter edge 6 (source destination weight): 1 2 3
Enter edge 7 (source destination weight): 1 7 3
Enter edge 8 (source destination weight): 2 3 2
Enter edge 9 (source destination weight): 2 7 5
Enter edge 10 (source destination weight): 3 6 4
Enter edge 11 (source destination weight): 3 4 7
Enter edge 12 (source destination weight): 4 5 6
Enter edge 13 (source destination weight): 4 7 6
Enter edge 14 (source destination weight): 5 6 4
Enter edge 15 (source destination weight): 6 7 2

Edge    Weight
7 - 1   3
1 - 2   3
2 - 3   2
0 - 4   3
0 - 5   3
5 - 6   4
6 - 7   2

Total Cost of MST: 20

Enter 'Y' for new session: n
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> g++ .\kruskal_min_tree.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> .\kruskal_min_tree.exe
Enter number of vertices and edges: 8 15
Edge 1 (src dest weight): 0 5 3
Edge 2 (src dest weight): 0 4 3
Edge 3 (src dest weight): 0 6 5
Edge 4 (src dest weight): 0 1 6
Edge 5 (src dest weight): 0 7 8
Edge 6 (src dest weight): 1 2 3
Edge 7 (src dest weight): 1 7 3
Edge 8 (src dest weight): 2 3 2
Edge 9 (src dest weight): 2 7 5
Edge 10 (src dest weight): 3 6 4
Edge 11 (src dest weight): 3 4 7
Edge 12 (src dest weight): 4 5 6
Edge 13 (src dest weight): 4 7 6
Edge 14 (src dest weight): 5 6 4
Edge 15 (src dest weight): 6 7 2

Edges in the Minimum Spanning Tree:
2 -- 3 == 2
6 -- 7 == 2
0 -- 4 == 3
1 -- 7 == 3
1 -- 2 == 3
0 -- 5 == 3
5 -- 6 == 4
Total Weight of MST: 20

Enter 'Y' for new session: n
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree>
```

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```
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> g++ .\dijkstra_short
path.c
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree> .\dijkstra_shortpath
.exe

Enter number of vertices: 8
Enter number of edges: 15
Edge 1 (src dest weight): 0 5 3
Edge 2 (src dest weight): 0 4 3
Edge 3 (src dest weight): 0 6 5
Edge 4 (src dest weight): 0 1 6
Edge 5 (src dest weight): 0 7 8
Edge 6 (src dest weight): 1 2 3
Edge 7 (src dest weight): 1 7 3
Edge 8 (src dest weight): 2 3 2
Edge 9 (src dest weight): 2 7 5
Edge 10 (src dest weight): 3 6 4
Edge 11 (src dest weight): 3 4 7
Edge 12 (src dest weight): 4 5 6
Edge 13 (src dest weight): 4 7 6
Edge 14 (src dest weight): 5 6 4
Edge 15 (src dest weight): 6 7 2
Enter the starting node for Dijkstra: 1

Vertex   Distance from Source (1)
0        INF
1        0
2        3
3        5
4        12
5        18
6        9
7        3

Enter 'Y' for new session: n
PS C:\Users\Sujal Shrestha\OneDrive\Documents\DSA\Tree>
```

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